

Reading waves

Waveforms plot each watched signal's value over simulation time

Starter D flip-flop

- Starter preset
- Watch clk, d, and q
- Poke D equals one, then click posedge clk
- Q rises on the wave at the clock edge
- Before the edge you can poke D equals one while q still reads zero
- Click inside the wave panel to move the cursor and confirm q at the cursor time

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Waveform lab

Grow a live digital wave from the HDL engine: step or take clock edges, click a time cursor, and read signal values. Starter: D flip-flop — poke D=1, then ↗posedge and watch q rise on the wave.

Starter example: D flip-flop with forever #5 clock. Poke D=1, ↗posedge, then click the wave to read values at the cursor.

Load starter example

Challenges 0/22

Pick one

Quiz: waveform

A digital waveform plot shows...

☐ each watched signal's value over simulation time

☐ only the Verilog source text

☐ place-and-route congestion

☐ analog SPICE currents only

Start

Show hint

Check

Next

Stop

Idle

Waveform lab starter

Real RTL/TB practice

- In Track A, sketch clk, d, and q for one posedge where D was one before the edge
- Write why q updates at the edge, not when d changes alone
- After a UART sim, open a VCD in your viewer and name three signals you would watch first
- This lab teaches wave literacy

Pitfalls to watch

- Do not treat eyeballing the wave as the pass criterion
- Do not assume q follows d instantly, posedge sampling and NBA timing matter
- A flat q with enable off is correct, not a broken sim
- Bus labels are binary or hex, read width before comparing
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load D-FF starter, poke D equals one, posedge clk until q equals one
- On paper, draw one clk period and mark where q would update
- When you are ready, take the short quiz, then continue to TB versus UVM map